I realize that this is the kind of hopelessly abstract statement that makes people give up on quantum mechanics and take up surfing instead. Don't worry—its meaning will become clear by the end of the lecture.

We'll soon see that **L** must also be Hermitian. Some authors regard this as a postulate, or basic principle. We have chosen instead to derive it from the other principles. The end result is the same either way: the operators that represent observables are Hermitian.

- Principle 2: The possible results of a measurement are the eigenvalues of the operator that represents the observable. We'll call these eigenvalues λ_i. The state for which the result of a measurement is unambiguously λ_i is the corresponding eigenvector $|\lambda_i\rangle$. Don't unpack your surfboard just yet.

  Here's another way to say it: if the system is in the eigenstate $|\lambda_i\rangle$, the result of a measurement is *guaranteed* to be λ_i.

- Principle 3: Unambiguously distinguishable states are represented by orthogonal vectors.

- Principle 4: If $|A\rangle$ is the state-vector of a system, and the observable **L** is measured, the probability to observe value λ_i is

$$P(\lambda_i) = \langle A|\lambda_i\rangle\langle\lambda_i|A\rangle. \qquad (3.11)$$

  I'll remind you that the λ_i are the eigenvalues of **L**, and $|\lambda_i\rangle$ are the corresponding eigenvectors.

These brief statements are hardly self-explanatory, and we'll need to flesh them out. For the moment, let's accept the first item, namely that every